What cross-species interactions can tell us about the communication of emotion

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1. Emotional vocalizations

Humans use language for many purposes, but one that is often discussed is the sharing or hiding of emotions between social partners, making emotional communication a good candidate for a driver of the evolution of language. While the study of negative emotions in nonhuman animals has been a topic of interest for decades, the study of positive affect in nonhuman animals has only recently been approached (Nelson et al., 2023). Initial studies focused on laughter or laughter-like vocalizations in rodents, apes, and parrots (see review in Winkler & Bryant, 2021) and showed that animals across a wide range of taxa display specific vocalizations during play-like behavior, supporting the interpretation that those behaviors are signaling positive affect.

1.1. Dolphins and positive affect

Until recently, no candidate vocalization for positive affect has been suggested for dolphins, however Ridgeway and colleagues (e.g. Dibble et al., 2016) reported that dolphins made a "victory squeal" – a specific kind of burst pulse (BP) vocalization just after their trainers "bridged" a behavior. A bridge is a secondary reinforcer – often a whistle – that serves as a marker that the dolphin has performed a correct behavior. Ridgeway therefore speculated that this victory squeal was likely an expression of positive affect. However, the acoustic characteristics of this vocalization were vaguely defined (although, see Jones et al., 2020). Recent work from our collaborators separated BPs into several

subtypes based on modulation patterns of click repetition rate and associated separate subtypes with emotional contexts (Jenks, 2025). Using this novel framework, they associated BPs with increasing click repetition rates (up) and those with increasing followed by decreasing patterns (up/down) with contexts of positive affect (Jenks, 2025, Lameris et al., in review).

1.2. Findings

We recently observed 5 bottlenose dolphins (*Tursiops truncatus*) during training sessions (on-session) and when receiving an unexpected object outside of training (off-session) at the Dolphin Research Center in Marathon, FL. We hoped to delineate the contexts in which different kinds of BPs were made and how they interacted with energy level, reinforcement, and social interaction. We recorded dolphin vocalizations using an underwater hydrophone array with an attached Zoom H6 Handy Recorder with an XYH-6 X/Y capsule. To capture interactions with objects and trainers, we used GoPros setup at 3 different camera angles. Thirty-one (31) off-session and 33 on-session trials were included in the analysis, with 3-8 sessions with each dolphin. We used the same subtype definitions as our colleagues and coded the trials sequentially via video and acoustic spectrograms. All behaviors were time-stamped to preserve the temporal order of trainer behaviors and dolphin responses.

During off-session trials, in the 10 seconds following an unexpected reward, we found that dolphins most often emit "up" BPs that were similar to those previously seen in positive affect contexts. For on-session trials, we performed a sequential analysis to identify the behaviors that occurred directly following a bridge. The first behavior to follow a bridge was predominantly a variation of the up/down BP that we've termed an up/down scream (fewer sidebands, but same modulation pattern, found 68.2% of the time). Similarly, the dolphins emitted the up/down scream variant 71.5% of the time following social interactions from their trainers such as cheering, pointing, and excited movements.

Taken together, our data suggest that dolphins may use an up/down BP vocalization as an indicator of positive affect and that this vocalization may be modulated by interaction with human trainers. The existence of social/emotional vocalizations in such distantly related species as humans, apes, dolphins, and parrots support the evolutionary importance of such signals, suggesting that sharing of positive affect/positive intentions may be an adaptation that may ultimately lead to complex communication, like language.

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